

Learning-based Control for Underwater Vehicles under Simulated Wave-Induced Disturbances

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Abstract: This paper presents a learning-based control strategy for marine vehicles station keeping under stochastic ocean disturbances modeled by the JONSWAP spectrum. A Deep Reinforcement Learning (DRL) policy, trained via Proximal Policy Optimization (PPO), is compared against PID and Model Predictive Control (MPC) benchmarks. Results demonstrate that the PID controller fails to provide acceptable tracking precision in both calm and rough sea states, being unable to compensate for the coupled hydrodynamic disturbances. While MPC excels in calm waters, the PPO policy provides superior robustness in storm conditions, achieving a 42.5% reduction in RMSE compared to MPC. Despite a higher absolute power demand than the classical baselines, PPO exhibits a three fold improvement in efficiency ratio over MPC in severe states. Furthermore, computational analysis shows the PPO inference is 104 times faster than the MPC optimization, confirming its suitability for real time deployment in resource constrained underwater vehicles.

Keywords: Marine vehicles, Learning-based control, Reinforcement learning, Trajectory tracking, Disturbance rejection.

1. INTRODUCTION

Industrial ocean operations require continuous inspection and monitoring of subsea assets over large areas. Marine vehicles and underwater robots play a central role in these tasks, as many are unsafe or impractical for human intervention (Nauert and Kampmann, 2023; Zhong et al., 2025; Paull et al., 2013). In practice, these platforms face distinct operational constraints. Autonomous Underwater Vehicles (AUVs) operate untethered, making energy-related control effort critical to mission success (Wynn et al., 2014). Remotely Operated Vehicles (ROVs), typically tethered, introduce constraints related to actuation limits and safe maneuvering (Christ and Wernli Sr, 2013). These differences motivate control strategies that explicitly account for platform-dependent constraints.

Across these applications, motion control is consistently challenged by ocean disturbances. Currents and wave-induced effects introduce time-varying external loads that degrade position tracking. Disturbance-tested studies consistently report that maintaining bounded errors under such variability remains a primary limiting factor for station keeping operations (Cao et al., 2020; Walker et al., 2021).

To address these challenges in station keeping and dynamic positioning, various control approaches have been established. Robust nonlinear controllers and observer-based architectures are frequently evaluated under strong currents and hydrodynamic uncertainty. For example, sliding mode architectures incorporating current observers (Mu et al.,

2021) have been proposed to reject unknown environmental loads.

Optimization-based formulations like Model Predictive Control (MPC) are widely applied when explicit constraint handling and anticipatory behavior are required. For instance, disturbance-tested NMPC designs incorporate explicit generalized force models for dynamic positioning (Cao et al., 2020). Expanding on anticipatory capabilities, recent wave-preview NMPC formulations integrate forecasted wave elevation data from upstream sensors to optimize control sequences against impending hydrodynamic loads (Walker et al., 2025). However, while these approaches offer structured disturbance mitigation, their interaction with realistic stochastic wave models is rarely addressed, often relying on simplified scenarios (Dai et al., 2025).

More recently, learning-based control has emerged as a promising alternative to reduce dependence on accurate hydrodynamic modeling. Reinforcement learning (RL) methods are applied to adapt control parameters and improve robustness under uncertainty (Lu et al., 2023). Comparative evidence suggests that learning-based controllers are often most effective as adaptation layers. While classical controllers like PID may achieve higher precision in specific station keeping tasks, learning-based methods demonstrate significant potential to reduce control effort and improve adaptability (Sola et al., 2020). These complementary strengths motivate hybrid architectures integrating learning with model-based controllers such as MPC (Cai et al., 2024).

Despite the breadth of these established and emerging control strategies, a critical limitation persists in how ocean disturbances are represented during controller evaluation. The literature highlights that many studies rely on simplified or deterministic disturbance models (Tugal et al., 2022; Walker et al., 2021). This limitation directly impacts the generalization of both model-based and learning-based controllers. In practice, ocean waves are stochastic processes that can be more accurately described using spectral models such as the JONSWAP spectrum, which characterizes energy distribution across frequencies. However, the integration of physically consistent wave models into control design and learning-based training frameworks remains limited.

Motivated by these gaps, this work investigates a learning-based low-level control framework for underwater robots focused on station keeping under realistic ocean disturbances. The proposed approach leverages Deep Reinforcement Learning (DRL), where control policies are trained in simulation using a dynamic model of the vehicle combined with stochastic disturbances generated from the JONSWAP spectrum. By exposing the controller to realistic disturbance variability during training, the approach aims to improve robustness and generalization.

The evaluation focuses on station keeping performance under disturbance-rich environments, considering tracking accuracy, disturbance rejection capability, and energy-related control effort, particularly relevant for AUV operations (Sola et al., 2020). The proposed DRL-based controller is quantitatively compared against established baselines, including PID and MPC, under consistent and physically meaningful disturbance scenarios.

The main contributions of this work are:

- The design and training of a DRL-based controller for underwater station keeping under wave-induced disturbances modeled via the JONSWAP spectrum.
- An evaluation of the controller’s capability to handle physically consistent stochastic disturbances while maintaining tracking performance.
- A comparative analysis between the proposed learning-based approach and classical control strategies (PID and MPC) under consistent disturbance conditions.
- A quantitative assessment of the trade-offs between tracking accuracy, control effort and computational cost across different sea states.

2. SYSTEM MODELING AND PROBLEM FORMULATION

2.1 Vehicle Dynamics

The BlueROV2 platform is modeled as a rigid-body underwater vehicle with six degrees of freedom (6-DOF), following a control-oriented formulation based on the Fossen (2011).

The generalized position and velocity vectors are defined as

$$\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^T, \quad \boldsymbol{\nu} = [u, v, w, p, q, r]^T, \quad (1)$$

where (x, y, z) denote position in the inertial frame, (ϕ, θ, ψ) are roll, pitch, and yaw angles, and (u, v, w, p, q, r) are the body-fixed linear and angular velocities.

The kinematics are given by

$$\dot{\boldsymbol{\eta}} = \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu}, \quad (2)$$

where $\mathbf{J}(\boldsymbol{\eta})$ is the transformation matrix between body-fixed and inertial frames.

The vehicle dynamics are described by

$$\mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}(\boldsymbol{\nu}, \boldsymbol{\nu}_r)\boldsymbol{\nu} + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) = \boldsymbol{\tau}, \quad (3)$$

where $\mathbf{M}$ is the inertia matrix, $\mathbf{C}(\cdot)$ is the Coriolis and centripetal matrix, $\mathbf{D}(\cdot)$ represents hydrodynamic damping, $\mathbf{g}(\boldsymbol{\eta})$ accounts for restoring forces, and $\boldsymbol{\tau}$ is the control input vector.

The inertia matrix is modeled as diagonal, combining rigid-body and added mass effects:

$$\mathbf{M} = \text{diag}(m, m, m, I_x, I_y, I_z) + \text{diag}(\mathbf{m}_A), \quad (4)$$

where m is the vehicle mass, (I_x, I_y, I_z) are the moments of inertia, and $\mathbf{m}_A = [m_{A,u}, m_{A,v}, m_{A,w}, m_{A,p}, m_{A,q}, m_{A,r}]$ is the added mass vector.

The relative velocity with respect to the surrounding fluid is defined as

$$\boldsymbol{\nu}_r = \boldsymbol{\nu} - [u_c, v_c, w_c, 0, 0, 0]^T, \quad (5)$$

where (u_c, v_c, w_c) are the current-induced velocity components. In this work, environmental disturbances act only on translational velocities.

Hydrodynamic damping is modeled as

$$\mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r = \mathbf{d}_L \odot \boldsymbol{\nu}_r + \mathbf{d}_Q \odot \boldsymbol{\nu}_r \odot |\boldsymbol{\nu}_r|, \quad (6)$$

where $\mathbf{d}_L$ and $\mathbf{d}_Q$ are linear and quadratic damping coefficient vectors.

The system is written in explicit form as

$$\dot{\boldsymbol{\nu}} = \mathbf{M}^{-1}(\boldsymbol{\tau} - \mathbf{C}(\boldsymbol{\nu}, \boldsymbol{\nu}_r)\boldsymbol{\nu} - \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r - \mathbf{g}(\boldsymbol{\eta})). \quad (7)$$

The system is discretized using an explicit Euler scheme with sampling time $\Delta t = 0.1$ s.

The control objective for station keeping is defined as

$$\boldsymbol{\eta}(t) \rightarrow \boldsymbol{\eta}_{ref}, \quad \boldsymbol{\nu}(t) \rightarrow \mathbf{0}. \quad (8)$$

The physical and hydrodynamic parameters used in the model are summarized in Table 1.

Table 1. Vehicle model parameters.

Symbol	Value
m	11.5 kg
V	0.0113459 m ³
ρ	1000 kg/m ³
g	9.81 m/s ²
d_{GB}	0.01 m
I_x, I_y, I_z	0.16 kg·m ²
$\mathbf{m}_A$	[5.50, 12.70, 14.57, 0.12, 0.12, 0.12]
$\mathbf{d}_L$	[4.03, 6.22, 5.18, 0.07, 0.07, 0.07]
$\mathbf{d}_Q$	[18.18, 21.66, 36.99, 1.55, 1.55, 1.55]

2.2 Wave Modeling via JONSWAP Spectrum

Ocean disturbances are modeled as wave-induced orbital velocities derived from the JONSWAP spectrum (Hasselmann et al., 1973), providing a physically consistent representation of irregular sea states.

The wave energy spectrum is defined in terms of angular frequency as

$$S(\omega) = \frac{\alpha g^2}{\omega^5} \exp \left[-\frac{5}{4} \left(\frac{\omega_p}{\omega} \right)^4 \right] \gamma \exp \left[-\frac{(\omega - \omega_p)^2}{2\sigma^2 \omega_p^2} \right], \quad (9)$$

where ω is the angular frequency, $\omega_p = 2\pi/T_p$ is the peak frequency, g is the gravitational acceleration, γ is the peak enhancement factor, σ is the spectral width parameter, and α is a scaling coefficient determined from the sea state parameters H_s (significant wave height) and T_p (peak period).

The spectrum is discretized into N frequency components, and the amplitude of each component is computed as

$$a_i = \sqrt{2S(\omega_i)\Delta\omega}, \quad (10)$$

where ω_i is the discretized angular frequency and $\Delta\omega$ is the frequency resolution.

A time-domain realization of the horizontal wave-induced velocity is obtained as

$$u_c(t) = \sum_{i=1}^N a_i \cos(\omega_i t + \phi_i), \quad (11)$$

where $\phi_i \sim \mathcal{U}(0, 2\pi)$ are independent random phases.

The disturbance is projected onto the horizontal plane as

$$\boldsymbol{\nu}_{wave}(t) = \begin{bmatrix} u_c(t)d_x \\ u_c(t)d_y \\ 0 \end{bmatrix}, \quad (12)$$

where (d_x, d_y) defines the wave propagation direction. This velocity is incorporated into the vehicle model as a current disturbance acting on the translational components of the relative velocity vector.

Two sea states are considered, representing mild and severe conditions. The corresponding parameters are summarized in Table 2.

Table 2. JONSWAP parameters for the evaluated sea states.

Parameter	Symbol	Calm	Storm
Significant wave height	H_s	1.0 m	5.5 m
Peak period	T_p	6.0 s	11.0 s
Peak enhancement factor	γ	3.3	4.5
Number of components	N	64	64
Wave direction	(d_x, d_y)	(0.5, 0.5)	(0.7, 0.7)

The PPO training hyperparameters used in this work are summarized in Table 3.

Table 3. PPO training hyperparameters used for the BlueROV control task

Parameter	Value
Total timesteps	1×10^6
Network architecture	2 layers (256 neurons each)
Learning rate	3×10^{-4}
Batch size	64
n_steps	2048
n_epochs	10
Discount factor γ	0.99
GAE λ	0.95
Clip range	0.2

3. METHODOLOGY

This section describes the experimental protocol and the methods used to evaluate tracking performance, energy consumption, and computational cost of the PID, MPC, and learning-based (PPO) controllers. The benchmark PID controller follows the cascaded architecture proposed in (Kermorgant, 2024). The predictive controller is formulated as a MPC scheme, whose internal model and optimization parameters are detailed in Appendix A.

3.1 Experimental Protocol and Duration

The controllers are evaluated under the sea states defined in Table 2. Based on empirical observations of the settling time under different disturbance intensities, the simulation duration T is adjusted for each environment:

- Calm Sea: $T = 40$ s (400 steps with $\Delta t = 0.1$ s)
- Rough Sea: $T = 80$ s (800 steps with $\Delta t = 0.1$ s)

Six station-keeping scenarios are considered. Each scenario starts from the same initial condition $\boldsymbol{\eta}_{init} = [0, 0, 0]^T$, and the vehicle is commanded to reach a fixed target position $\boldsymbol{\eta}_{target}$. The setpoints are listed in Table 4, covering both single-axis and coupled multi-axis stabilization.

Table 4. Station-keeping target setpoints.

Scenario ID	Target Position $[x, y, z]^T$ [m]
1	[0.0, 0.0, -0.5]
2	[0.5, 0.0, -0.5]
3	[-0.5, 0.0, -0.5]
4	[0.0, 0.5, -0.5]
5	[0.0, -0.5, -0.5]
6	[0.5, 0.5, -0.7]

3.2 Performance and Energy Metrics

Tracking performance is evaluated using the Root Mean Square Error (RMSE), computed over N samples as

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{k=1}^N \|\mathbf{p}(k) - \mathbf{p}_d\|^2} \quad (13)$$

where $\mathbf{p}(k)$ is the vehicle position at time step k , $\mathbf{p}_d$ is the target position, and N is the total number of samples.

The steady-state error is estimated as the average tracking error over the final 5 seconds of each simulation.

Energy consumption is modeled based on the BlueROV2 T200 thrusters. The instantaneous power of each thruster is computed as

$$P_i(k) = P_{max} \left(\frac{|f_i(k)|}{f_{max}} \right)^{1.5} \quad (14)$$

where $f_i(k)$ is the force produced by the i -th thruster at time step k , $f_{max} = 50$ N is the maximum thrust, and $P_{max} = 350$ W is the maximum power.

The total power and cumulative energy are defined as

$$P(k) = \sum_{i=1}^6 P_i(k), \quad E_{cum} = \sum_{k=1}^N P(k) \Delta t \quad (15)$$

where $P(k)$ is the total power at time step k and Δt is the simulation time step.

The average power consumption is given by

$$\bar{P} = \frac{1}{N} \sum_{k=1}^N P(k) \quad (16)$$

For each scenario, RMSE, cumulative energy, and average power are computed independently and aggregated across scenarios using mean and standard deviation.

To quantify the trade-off between tracking accuracy and power consumption, an efficiency ratio is defined as

$$\rho = \frac{\text{RMSE}}{\bar{P}} \quad (17)$$

where lower values of ρ indicate improved tracking performance relative to the required power.

3.3 Computational Performance Evaluation

The computational performance of each controller is evaluated by measuring both wall-clock execution time and CPU processing time at each control step k .

The wall time $t_{\text{wall}}(k)$ represents the real elapsed time required to compute the control action, while the CPU time $t_{\text{cpu}}(k)$ corresponds to the effective processing time consumed by the controller.

The simulation time step $\Delta t = 0.1$ s defines the real-time constraint. A deadline miss occurs whenever the execution time exceeds the available control interval, i.e.,

$$t_{\text{wall}}(k) > \Delta t, \quad f_{\text{eq}}(k) = \frac{1}{t_{\text{wall}}(k)} \quad (18)$$

where $f_{\text{eq}}(k)$ represents the instantaneous control frequency associated with the wall-clock execution time.

For each scenario, statistical metrics are computed from the execution time series, including mean, standard deviation, maximum value, and 95th percentile.

The ratio between CPU time and wall time is computed as

$$\rho_{\text{cpu/wall}} = \frac{\sum_k t_{\text{cpu}}(k)}{\sum_k t_{\text{wall}}(k)} \quad (19)$$

which quantifies the proportion of time effectively spent in computation.

All computational metrics are first evaluated independently for each scenario and then aggregated across scenarios using mean values, yielding the summary results reported in Table 6.

4. DEEP REINFORCEMENT LEARNING CONTROL

The control problem is modeled as a Markov Decision Process (MDP), defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R})$, where $\mathcal{S}$ denotes the state space, $\mathcal{A}$ the action space, $\mathcal{P}$ the transition dynamics given by the conditional distribution $\mathcal{P}(\mathbf{s}_{t+1}|\mathbf{s}_t, \mathbf{a}_t)$, and $\mathcal{R}$ the reward function. At each time step t , the agent observes the system state $\mathbf{s}_t \in \mathcal{S}$ and selects an action $\mathbf{a}_t \in \mathcal{A}$ according to a stochastic policy $\pi(\mathbf{a}_t|\mathbf{s}_t)$, receiving a scalar reward $r_t \in \mathbb{R}$. The objective is to maximize the expected discounted cumulative reward:

$$\mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t r_t \right] \quad (20)$$

where $\gamma \in [0, 1]$ is the discount factor.

The overall control architecture is illustrated in Fig. 1.

The state vector is defined as:

$$\mathbf{s} = [x, y, z, \phi, \theta, \psi, u, v, w, p, q, r]^T \quad (21)$$

which includes the position vector $\mathbf{p} = [x, y, z]^T$ in the inertial frame, the orientation described by Euler angles ϕ (roll), θ (pitch), and ψ (yaw), the body-fixed linear velocity vector $\mathbf{v} = [u, v, w]^T$, and the angular velocity vector $\boldsymbol{\omega} = [p, q, r]^T$.

The action space consists of generalized forces and moments:

$$\mathbf{a} = [\tau_x, \tau_y, \tau_z, \tau_\phi, \tau_\theta, \tau_\psi]^T \quad (22)$$

where τ_x, τ_y, τ_z denote control forces along surge, sway, and heave, and $\tau_\phi, \tau_\theta, \tau_\psi$ denote control moments around roll, pitch, and yaw, corresponding to the net actuation generated by the thruster system.

The control inputs are bounded as:

$$\mathbf{a} \in [-50, 50]^6 \quad (23)$$

where the bounds define the admissible actuation limits.

The policy $\pi(\mathbf{a}|\mathbf{s})$ is represented by a parametric function approximator operating over continuous state and action spaces.

The reward function is designed to enforce station keeping performance while preserving system stability, and is defined as:

$$r = - \left(w_p \|\mathbf{p}\| + w_v \|\mathbf{v}\| + w_\psi |\tilde{\psi}| + w_s (|\phi| + |\theta|) \right) \quad (24)$$

where $\mathbf{p} = [x, y, z]$ denotes the position error relative to the reference, $\|\cdot\|$ is the Euclidean norm, $\mathbf{v} = [u, v, w]$ is the linear velocity vector, $\tilde{\psi}$ is the wrapped yaw error computed as the shortest angular difference in $[-\pi, \pi]$, ϕ and θ are roll and pitch angles, and $w_p, w_v, w_\psi, w_s > 0$ are scalar weighting coefficients.

This formulation penalizes deviation from the desired position, excessive velocities, orientation errors, and instability.

A small positive reward is assigned when the position error falls below a predefined threshold ϵ_p . Episodes are terminated when safety constraints are violated, such as exceeding position bounds or reaching unstable orientations, defined by limits on $\|\mathbf{p}\|$, $|\phi|$, and $|\theta|$.

5. RESULTS AND DISCUSSION

This section presents the comparative performance of the PID, MPC, and PPO controllers. The evaluation is divided into the convergence of the learning process, steady-state performance in calm and storm sea, and computational cost. All metrics are based on tracking precision and computational efficiency, as well as the power consumption derived from the T200 thruster models.

5.1 Policy Training and Convergence

The training metrics provide evidence that the neural network successfully internalized the control objectives defined by the reward function. Figure 2 presents the evolution of the accumulated reward over the training episodes.

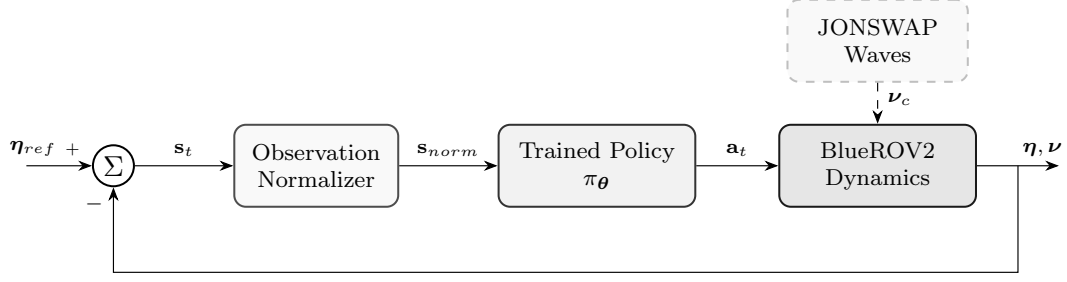


Figure 1. Proposed DRL-based control architecture for AUV station keeping: the trained PPO policy π_θ maps the normalized observation vector s_{norm} to generalized control actions a_t , where θ denotes the trainable parameters of the policy. The system explicitly accounts for wave-induced orbital velocities ν_c , with $\nu_c = [u_c, v_c, w_c, 0, 0, 0]^T$ representing the fluid velocity components in the body-fixed frame, within the BlueROV2 dynamics.

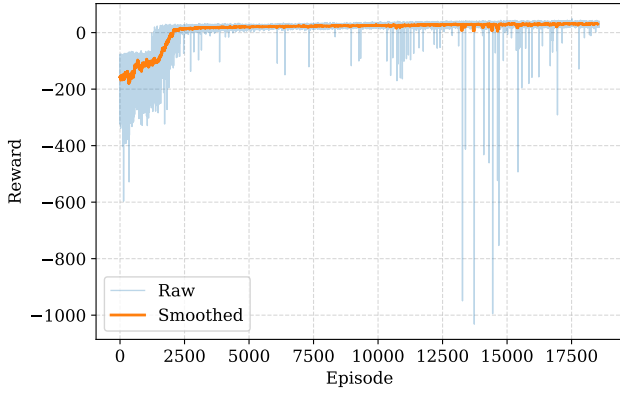


Figure 2. Evolution of the accumulated reward during the training phase.

The consistent increase and subsequent plateau of the reward indicate that the policy converged to a stable solution. This behavior suggests that the agent learned to balance the competing objectives of position regulation and energy economy. The convergence of the tracking error during training, illustrated in Figure 3, corroborates this analysis.

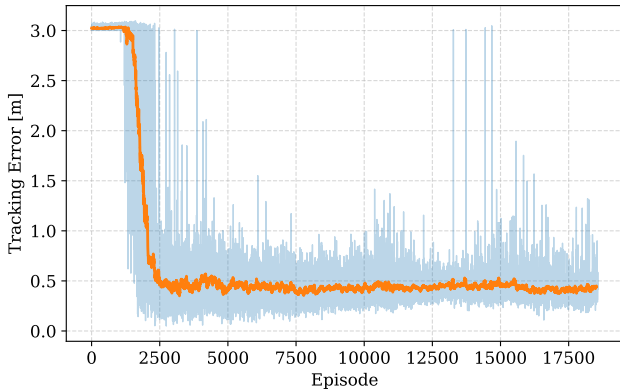


Figure 3. Reduction of the position tracking error over training episodes.

5.2 Comparative Performance Analysis

The tracking precision and energy efficiency for all evaluated controllers are summarized in Table 5. These metrics,

aggregated across 12 different scenarios, provide a quantitative basis for discussing the trade-offs between classical, predictive, and learning-based control strategies.

Table 5. Tracking performance, energy consumption, and efficiency ratio (ρ).

Scenario	Controller	Mean RMSE $\pm$ Std (m)	Avg. Power $\pm$ Std (W)	Efficiency ρ (m/W)
Calm Sea	PID	0.2702 ± 0.0289	1.56 ± 0.05	0.1732
	MPC	0.1387 ± 0.0214	2.78 ± 0.42	0.0499
	PPO	0.1778 ± 0.0428	4.38 ± 0.79	0.0406
Rough Sea	PID	1.1147 ± 0.0948	4.30 ± 0.08	0.2592
	MPC	0.2854 ± 0.0070	11.69 ± 0.21	0.0244
	PPO	0.1640 ± 0.0174	19.60 ± 0.59	0.0084

Note: Lower values of ρ indicate higher control efficiency.

In calm sea states, the evaluation reveals distinct transient and steady-state behaviors. As illustrated in Figure 4, the MPC achieves the fastest convergence, reaching a steady-state error of less than 0.1 meters in approximately 3 seconds. According to Table 5, the MPC maintains the lowest RMSE (0.1387 ± 0.0214 m). In terms of energy consumption, the PPO controller requires higher average power (4.38 ± 0.79 W) compared to the MPC (2.78 ± 0.42 W), suggesting a less efficient control action under low-disturbance conditions.

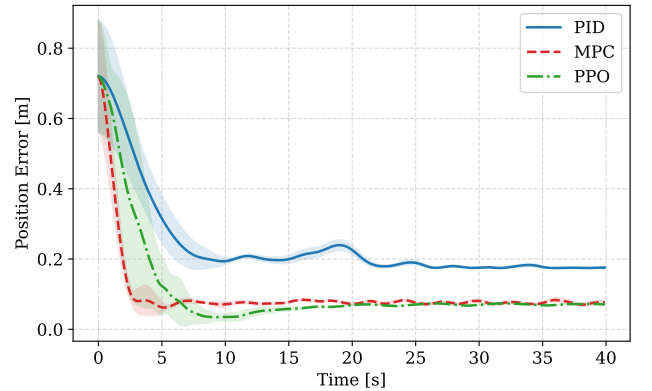


Figure 4. Positioning error performance under calm sea states.

The performance of the controllers was tested under rough sea states. As depicted in Figure 5, the wave-induced orbital velocities expose the limitations of classical approaches. The PID controller fails to maintain station-keeping, with the RMSE degrading to 1.1147 ± 0.0948 m.

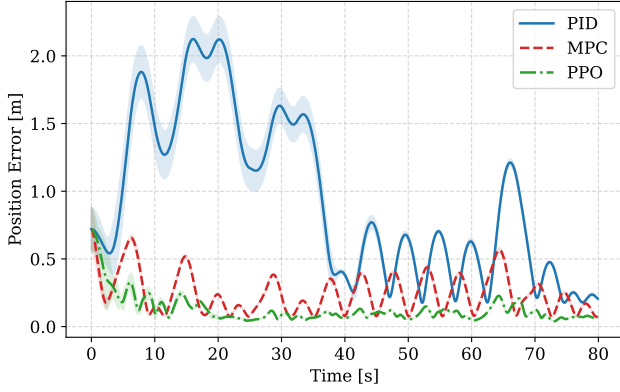


Figure 5. Positioning error performance under severe storm sea states.

The PPO controller demonstrates superior disturbance rejection, achieving the lowest mean RMSE (0.1640 ± 0.0174 m). According to Table 5, this performance is achieved at the cost of significantly higher average power consumption (19.60 ± 0.59 W). Despite this, the efficiency ratio $\rho = 0.0084$ m/W is significantly better than its counterparts. This justifies the energy investment, as the DRL-based policy provides a level of stability and consistency that classical methods cannot reach under severe stochastic disturbances.

5.3 Computational Requirements and Real-time Feasibility

A comprehensive computational analysis was conducted to assess the real-time viability of the controllers, as presented in Table 6. The metrics were aggregated over 6 scenarios with 800 steps each, totaling 4,800 control cycles per controller, executed on a system equipped with an NVIDIA RTX 4050 GPU and an Intel Core i7 processor.

Table 6. Detailed computational performance and real-time feasibility metrics.

Metric	PID	PPO	MPC
Mean Wall Time (ms)	0.0468	0.3611	37.5830
Wall Time Std. Dev. (ms)	0.0094	0.1563	0.8930
95th Percentile (ms)	0.0509	0.5077	43.7838
Max Wall Time (ms)	0.2020	1.5444	111.8352
Equiv. Processing Rate (Hz)	22,299.1	3,203.5	28.2
CPU to Wall Ratio	0.9857	0.9903	0.9996
Deadline Miss Rate (%)	0.00	0.00	0.52

Note: These parameters were extracted from the rough sea state scenario (Scenario 2).

The MPC controller exhibited the highest computational burden, with a mean execution time of 37.58 ms. A critical observation is the maximum wall time of 111.84 ms and a deadline miss rate of 0.52%. Given that the simulation sampling interval is $\Delta t = 100$ ms, these spikes indicate that the SLSQP-based optimization occasionally exceeds the real-time budget, which may compromise control stability in high-dynamic maneuvers. The CPU-to-Wall ratio of 0.9996 confirms that the execution is almost entirely bound by processing tasks, with negligible I/O overhead.

In contrast, the learning-based PPO policy demonstrated high computational efficiency, maintaining a mean wall

time of 0.3611 ms, approximately 104 times faster than the MPC. Even its 95th percentile (0.5077 ms) remains well within safety margins, providing an equivalent processing rate of 3,203.5 Hz. This efficiency stems from the policy’s implementation as a feed-forward neural network, which replaces iterative optimization with deterministic matrix operations.

The PID controller remained the most efficient baseline (0.0468 ms), with zero deadline misses. However, when comparing the trade-off between disturbance rejection and computational cost, the PPO agent offers a superior profile. It provides a good performance on station-keeping capabilities under storm conditions with a negligible increase in computational footprint compared to the classical PID, while avoiding the potential real-time violations observed in the MPC formulation.

The temporal distribution of these requirements is further analyzed in Figure 6, which illustrates the CPU time evolution throughout the simulation scenarios.

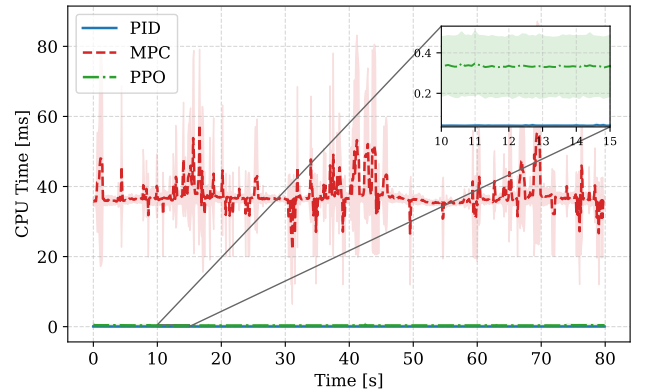


Figure 6. CPU time for each controller in experiment 2 under storm ocean.

6. CONCLUSION

This work investigated the application of Deep Reinforcement Learning for station keeping of AUVs under stochastic ocean disturbances modeled via the JONSWAP spectrum. The results showed that the PPO-based controller achieves superior tracking performance under rough sea conditions, effectively handling nonlinear hydrodynamic effects and disturbance interactions that challenge classical and model-based approaches.

Compared to PID and MPC, the learning-based policy provided improved positioning accuracy and more consistent performance across scenarios, while maintaining real-time feasibility with a significantly lower computational cost than MPC. Although higher power consumption was observed, the proposed efficiency ratio ρ indicates a more favorable trade-off between tracking performance and energy usage in highly disturbed environments.

Future research will focus on enhancing orientation control through specialized reward structures to improve heading stability in 6 DOF maneuvers. Additionally, the framework will be expanded to handle dynamic trajectory tracking with time varying setpoints under severe disturbances. Finally, new investigations will target energy aware policies

using multi objective reinforcement learning formulations to further balance energy economy during low disturbance periods.

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Appendix A. MODEL PREDICTIVE CONTROL IMPLEMENTATION

The Model Predictive Control (MPC) strategy is implemented using a nonlinear receding horizon formulation. At each control step, a finite-horizon optimal control problem is solved based on the system dynamics, and only the first control input of the optimal sequence is applied.

The system state $\mathbf{x} \in \mathbb{R}^{12}$ and control input $\mathbf{u} \in \mathbb{R}^6$ follow the definitions in (21) and (22), respectively. The formulation assumes regulation to the origin, i.e., $\mathbf{x}_{ref} = \mathbf{0}$.

The control problem is formulated over a prediction horizon of length N as:

$$\min_{\mathbf{U}} \sum_{k=0}^{N-1} (\mathbf{x}_k^\top Q \mathbf{x}_k + \mathbf{u}_k^\top R \mathbf{u}_k + \Delta \mathbf{u}_k^\top S \Delta \mathbf{u}_k) \quad (\text{A.1})$$

where $\mathbf{U} = [\mathbf{u}_0^\top, \dots, \mathbf{u}_{N-1}^\top]^\top$ is the stacked optimization variable, $\mathbf{x}_k \in \mathbb{R}^{12}$ is the predicted state at step k , $\mathbf{u}_k \in \mathbb{R}^6$ is the predicted control input, $\Delta \mathbf{u}_k = \mathbf{u}_k - \mathbf{u}_{k-1}$ is the control increment, and $Q \in \mathbb{R}^{12 \times 12}$, $R \in \mathbb{R}^{6 \times 6}$, and $S \in \mathbb{R}^{6 \times 6}$ are diagonal positive semi-definite weighting matrices associated with state deviation, control effort, and control variation, respectively.

The optimization is subject to the dynamic and actuator constraints:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k), \quad \mathbf{u}_k \in [\mathbf{u}_{\min}, \mathbf{u}_{\max}], \quad k = 0, \dots, N-1 \quad (\text{A.2})$$

where $f(\cdot)$ denotes the discrete-time nonlinear vehicle dynamics obtained by explicit Euler integration with sampling time Δt , $\mathbf{u}_{\min}$ and $\mathbf{u}_{\max}$ are the lower and upper actuator bounds, and $\mathbf{u}_{k-1}$ denotes the control input applied at the previous step for the computation of $\Delta \mathbf{u}_k$.

The controller employs $N = 10$ and $\Delta t = 0.1$ s. The resulting nonlinear programming problem is solved using Sequential Least Squares Programming (SLSQP) within a receding horizon (RH) framework.